

Indoor Wayfinding System for Small Buildings Using ESP32 Checkpoint Nodes and Web-Based Guidance

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Abstract—Indoor navigation is still a practical problem in small buildings such as clinics, academic departments, service centers, and office blocks, where visitors often need simple and quick room guidance rather than full indoor localization. This paper presents an indoor wayfinding system based on ESP32 checkpoint nodes, a master ESP32 hotspot node, and dual user guidance interfaces through a touch display unit and a local web page. The current prototype implements fixed routing for two destinations, namely Room A and Room B, using the checkpoint path Entrance → Middle → Destination. At the entrance, the master ESP32 creates the local hotspot and communicates with an Arduino UNO carrying a 2.4 inch touch LCD shield for destination selection and direction display. The same route information can also be accessed from a phone connected to the master node through a browser-based local web interface. Bluetooth Low Energy (BLE) based checkpoint updates are triggered when distributed ESP32 nodes detect a predefined BLE service UUID from the user's phone above a chosen RSSI threshold and report the checkpoint to the master node over HTTP through the local Wi-Fi network. The proposed system is intended for practical indoor guidance in small buildings, with particular relevance to urgent service and emergency-room navigation.

Index Terms—indoor wayfinding, ESP32, Arduino UNO, BLE, Wi-Fi hotspot, local web interface, touch LCD shield, checkpoint navigation, emergency guidance

I. INTRODUCTION

Indoor navigation inside small buildings is often treated as a minor problem, but in practice it affects visitor convenience, service speed, and response time during urgent situations. In

hospitals, diagnostic centers, college departments, and office buildings, users usually do not need centimeter-level indoor positioning. They need simple, understandable, and dependable guidance from one decision point to another. This makes checkpoint-based navigation more suitable than full precise localization for many real deployments.

The proposed work is an Internet of Things (IoT) based indoor wayfinding system that uses ESP32 nodes placed at major checkpoints such as the entrance, middle junction, and destination rooms. The current prototype emphasizes practical deployment and user interaction and presently supports two destinations, Room A and Room B. At the entrance, a master ESP32 node hosts a local hotspot and exchanges route information with an Arduino UNO fitted with a 2.4 inch touch LCD shield. Through this display unit, the user can choose a destination and read directions directly. Alternatively, the same guidance can be accessed through a local web page opened on the user's phone after connecting to the hotspot. BLE-based checkpoint reporting is used to update the user's approximate route progress automatically.

This project is important in the present era of IoT because it demonstrates how connected embedded nodes, local wireless communication, and simple user interfaces can solve a real human-centered navigation problem without relying on expensive infrastructure. The work is especially relevant for public-service and emergency-oriented buildings where fast guidance can reduce confusion and delay.

II. PROBLEM STATEMENT

Visitors in small buildings often face difficulty in locating rooms, departments, laboratories, or emergency service points because indoor signage may be insufficient, unclear, or overlooked under stress. Existing indoor localization solutions are often designed for high precision and can become unnecessarily complex for buildings that only require room-level or checkpoint-level guidance. Therefore, there is a need for a practical indoor wayfinding system that can guide users through a known sequence of checkpoints using simple hardware, local wireless networking, and easy-to-understand directional output.

III. OBJECTIVES

The main objectives of the proposed project are as follows:

- To design a checkpoint-based indoor wayfinding system for small buildings.
- To provide destination selection and guidance directly on an entrance touch LCD shield interface.
- To provide the same guidance on a phone through a local web page served by the master node.
- To use a master ESP32 node to create the hotspot and coordinate the local network.
- To use BLE-based checkpoint reporting for rough automatic route updates.
- To guide users to rooms and service points without requiring precise indoor localization.
- To support urgent service and emergency-room guidance in a practical embedded system.

IV. RELATED WORK

Indoor navigation based on BLE beacons and smartphone guidance has been widely studied in recent years. Guide-Beacon demonstrated beacon-based indoor wayfinding for visually impaired and disoriented users, showing the importance of structured indoor guidance [4]. BLE-based pedestrian guidance has also been explored for turn-by-turn navigation in indoor parking environments [5]. More recent work has discussed BLE-supported indoor navigation in structured facilities such as hospitals and repeated floor layouts [6]. These studies show that BLE-assisted indoor guidance is technically practical, but many existing solutions assume broader deployment complexity or more advanced positioning logic.

V. NOVELTY AND SOCIAL RELEVANCE

The novelty of the present work lies in its practical small-building architecture rather than full precise localization accuracy. The proposed system combines the following features in one compact design:

- A master ESP32 that creates the local hotspot for both nodes and users.
- An entrance Arduino UNO with 2.4 inch touch LCD shield for direct destination selection and direction display.

- A browser-based local web interface that works over the same local network without depending on external internet access.
- Checkpoint-based BLE updates instead of continuous coordinate estimation.
- A layout suitable for room guidance in small clinics, offices, departments, and emergency service buildings.

This work can help society by improving accessibility and reducing confusion in indoor public spaces. It is especially helpful for first-time visitors, elderly users, patients, and attendants who may need rapid access to an emergency room, registration desk, consultation room, or laboratory.

VI. PROPOSED SOLUTION

The proposed system uses multiple checkpoint nodes in a small indoor environment. The current setup includes a master entrance node, a middle node placed near the internal junction, and destination checkpoints for Room A and Room B. The master node creates a hotspot to which the other nodes and the user's phone can connect. At the entrance, an Arduino UNO with a 2.4 inch touch LCD shield acts as the local display terminal and exchanges data with the master ESP32 over serial communication. When a user selects destination A or B on the touch display, or through the local web page, the master node sets the current node to Entrance and generates the first movement instruction.

The system does not attempt continuous precise localization. Instead, it tracks movement at the checkpoint level. Each distributed ESP32 checkpoint node scans for a predefined BLE service UUID from the tracked user phone. When that UUID is detected strongly enough, the node sends an HTTP report to the master ESP32 with the detected checkpoint name. The master node then updates the active route state and displays the next instruction either on the entrance display or on the phone interface.

VII. SYSTEM ARCHITECTURE

The system architecture consists of three main layers:

A. Master Node Layer

The entrance ESP32 acts as the master node. It creates the Wi-Fi hotspot, hosts the route-selection logic and local web interface, communicates with the Arduino UNO display controller over serial, and serves as the main interaction point for the user.

B. Checkpoint Node Layer

Additional ESP32 nodes are placed at the middle corridor junction and near destination rooms. These nodes represent checkpoints, scan for the user's BLE advertisement, and report node transitions to the master over HTTP through the hotspot network.

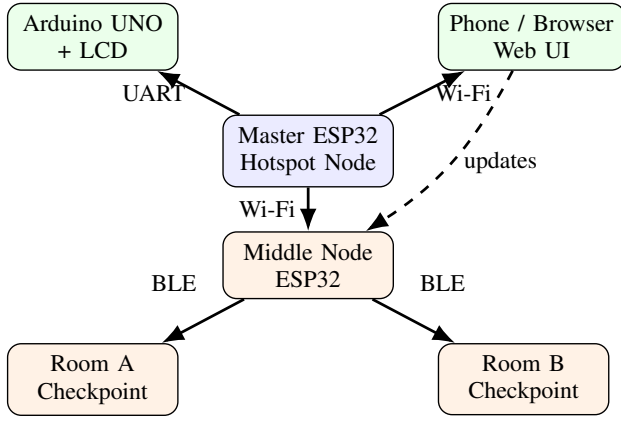


Fig. 1. System architecture of the prototype with the master ESP32, Arduino UNO plus LCD unit, web UI, and checkpoint nodes.

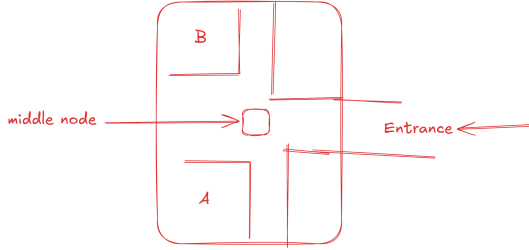


Fig. 2. Checkpoint layout of the experimental floor plan with the entrance, middle node, and room checkpoints A and B.

C. User Interface Layer

The user can interact with the system in two ways:

- By selecting the destination directly on the 2.4 inch touch LCD shield at the entrance.
- By connecting a smartphone to the master ESP32 hotspot and opening the local web page.

VIII. CURRENT EXPERIMENTAL LAYOUT

Based on the present prototype layout, the building entrance is placed on the right side of the corridor path, the middle checkpoint node is located near the central junction, and the destination rooms are represented by checkpoints such as Room A and Room B on the left side of the plan. This arrangement demonstrates how a user entering from the right side can be guided first to the middle checkpoint and then redirected to the selected room.

IX. HARDWARE COMPONENTS USED

Table I lists the main components used in the current prototype with current market prices in Indian Rupees.

The listed prices are practical current estimates for the hardware used in the prototype. The resistor-network cost is shown as an approximate value because such components are usually sold in small packs or mixed assortments.

TABLE I
HARDWARE COMPONENTS, KEY FEATURES, AND CURRENT PRICE

Component	Key Features	Role in Project	Price
ESP32 development board	Wi-Fi, BLE, GPIO support, easy prototyping	Master node and checkpoint node platform; price listed per board	Rs. 549.00
Arduino UNO	UART support, easy shield integration, simple display control	Entrance display controller linked with master ESP32	Rs. 549.00
2.4 inch touch LCD shield	Touch input and local visual output	Destination selection and on-device guidance at entrance	Rs. 475.00
Smartphone	Browser-based user access and BLE advertising support	Opens the local web page and carries the tracked BLE signal	User-provided
Resistor network	Logic-level adaptation and interface support	Used in the ESP32-UNO serial interconnection and voltage divider path	Rs. 12.00 (approx.)
Power supply / USB adapter	Stable node power	Powers ESP32 and display modules	Rs. 145.00

X. ROLE OF EACH COMPONENT

A. Master ESP32 Node

The master ESP32 at the entrance is the central node of the system. It creates the hotspot, coordinates communication, handles route selection requests, and exchanges route data with the Arduino UNO over serial communication.

B. Checkpoint ESP32 Nodes

The checkpoint ESP32 nodes are installed at the middle junction and near room checkpoints. They scan for the predefined BLE service UUID from the user phone, apply an RSSI threshold, and report confirmed checkpoint detection to the master ESP32.

C. Arduino UNO

The Arduino UNO is used as the local display controller at the entrance. The 2.4 inch touch LCD shield is mounted directly on top of the UNO, and the UNO exchanges serial data with the ESP32.

D. 2.4 Inch Touch LCD Shield

The touch LCD shield provides a local kiosk interface. The user can select the destination and read directions directly on the display without needing to use a phone.

E. Phone and Local Web Interface

The mobile phone connects to the master hotspot and opens the locally hosted web page. The same phone is also used as the BLE source device for checkpoint reporting by advertising the tracked BLE service UUID.

TABLE II
ESP32 AND UNO SERIAL CONNECTIONS USED IN THE PROTOTYPE

ESP32 Pin	Connected Module	Purpose
TX2	UNO RX	ESP32 sends route or control data to the Arduino UNO
RX2	UNO TX through voltage divider	ESP32 safely receives UART data from the 5 V UNO side
GND	UNO GND	Common ground reference between ESP32 and Arduino UNO

XI. METHODOLOGY

A. Approach for Prototyping

The prototype is developed as a room-guidance system rather than a full localization platform. The design starts by identifying important checkpoints inside the building and assigning one node to each major decision point. In the present implementation, the route logic is fixed for two destinations only: Entrance $\rightarrow$ Middle $\rightarrow$ A and Entrance $\rightarrow$ Middle $\rightarrow$ B.

B. Circuit and Node Description

In the current design, the 2.4 inch touch LCD shield is mounted directly on top of the Arduino UNO by inserting the shield header pins into the UNO female headers. The Arduino UNO acts as the display-side controller, while the master ESP32 handles hotspot creation, route logic, and HTTP server functions. The UNO and ESP32 are connected through UART links. The UNO sends user commands such as A, B, and reset to the master, while the master sends back a serial packet containing destination, current node, direction code, and instruction text. The other checkpoint nodes connect to the hotspot and send HTTP reports to the master when BLE detection succeeds. A simulation or pre-deployment circuit draft may also be prepared using Wokwi before final hardware assembly [7].

C. Important ESP32 and UNO Connections

Because the submission guidelines explicitly request the important used pins of ESP32, Table II records the main serial connections used in the present prototype.

The voltage divider is necessary on the UNO TX to ESP32 RX2 path because the Arduino UNO operates at 5 V logic, while the ESP32 uses 3.3 V logic levels.

D. Mathematical Expression for BLE-Based Rough Detection

The BLE checkpoint update is approximate and can be related to received signal strength indicator (RSSI) behavior. The node firmware uses an RSSI threshold check for reporting, and a commonly used path-loss form is

$$RSSI(d) = RSSI(d_0) - 10n \log_{10} \left(\frac{d}{d_0} \right), \quad (1)$$

where d is the estimated distance, d_0 is a reference distance, and n is the environment-dependent path-loss exponent. In this

project, the expression is used only as a rough interpretive model for checkpoint proximity, not as a precise indoor localization equation.

E. Dataset and AI/ML Methodology Note

The current prototype does not depend on any external dataset or AI/ML model. Route generation is based on pre-defined checkpoint paths, and checkpoint updates are based on BLE UUID detection, RSSI thresholding, and direct HTTP node reporting. If future versions adopt learned RSSI classification, user-movement prediction, or adaptive route optimization, then a dedicated dataset collection and AI/ML workflow can be added in this section.

XII. WORKING PRINCIPLE

The complete working flow is as follows:

- 1) The entrance master ESP32 powers up and creates the local hotspot.
- 2) The Arduino UNO initializes the touch LCD shield and shows the destination selection interface.
- 3) Checkpoint nodes connect to the master hotspot and start BLE scanning.
- 4) The user selects destination A or B either on the touch LCD shield or on the phone-based local web page.
- 5) The master node maps the selected destination to a fixed checkpoint sequence, for example Entrance $\rightarrow$ Middle $\rightarrow$ A.
- 6) The master sends destination, node state, direction code, and instruction text to the Arduino UNO over UART.
- 7) The first direction is shown on the touch LCD shield and, if required, on the phone web page.
- 8) As the user moves through the building, a checkpoint node detects the tracked BLE service UUID with sufficient RSSI and sends an HTTP report to the master node.
- 9) The master validates the reported node, updates the route state, and generates the next instruction.
- 10) The process continues until the destination checkpoint is reached.

XIII. SOFTWARE FLOW

The software logic can be summarized as follows:

- 1) Initialize the master ESP32 in hotspot mode.
- 2) Start the local HTTP server and local web page service on the master node.
- 3) Initialize the Arduino UNO and the touch LCD shield for destination selection and route display.
- 4) Accept user destination input from the display or from the browser-based web page.
- 5) Load the fixed route path for destination A or B.
- 6) Transmit the route state to the UNO as a serial packet.
- 7) Monitor HTTP reports from distributed nodes after BLE UUID detection.
- 8) Update the active checkpoint and push the next direction to the display and web page.

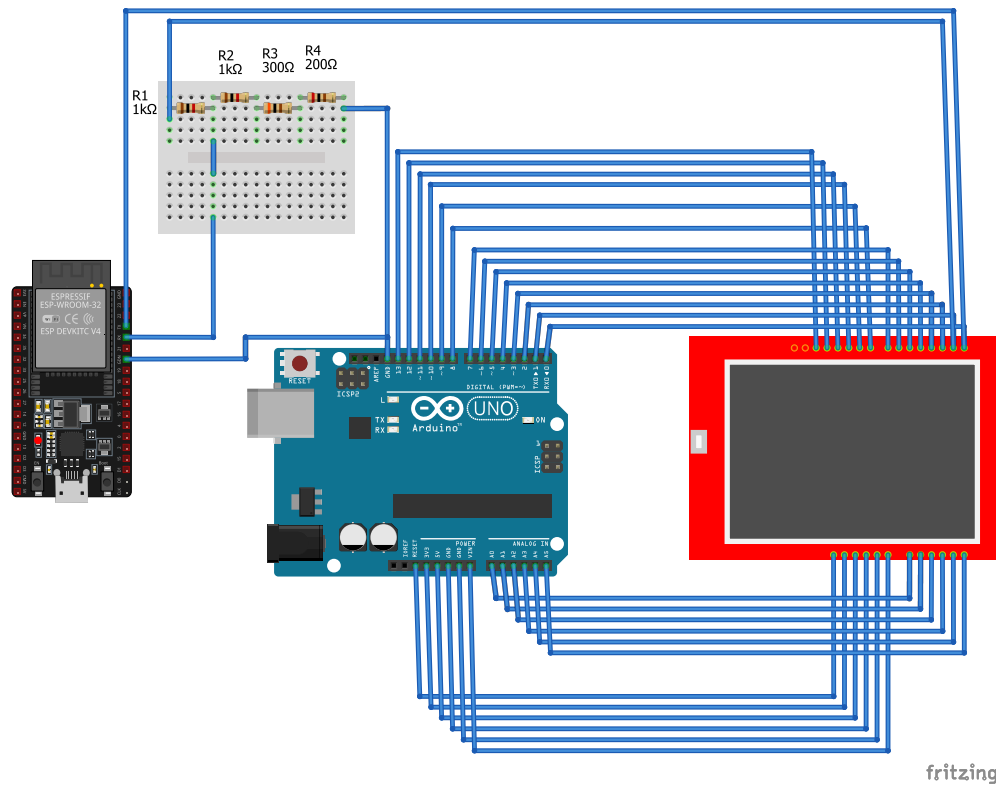


Fig. 3. Board-level wiring diagram of the prototype showing the ESP32 module, Arduino UNO, 2.4 inch LCD module, and resistor-based serial interface network.

- 9) Mark the destination as reached when the final checkpoint is confirmed.

XIV. REAL-TIME PRACTICAL EXPERIMENT, RESULTS, AND DISCUSSION

The present prototype can be experimentally demonstrated in a small corridor layout containing an entrance checkpoint, a middle node, and destination checkpoints A and B. A user enters from the entrance side, selects the destination, and then follows the displayed step-by-step directions. The experiment is considered successful when the user can reach the correct room using only the displayed route instructions and checkpoint updates. The implemented entrance display is shown in Fig. 6, while the browser-based local web interface is shown in Fig. 7.

The main expected observations are:

- The touch LCD interface allows destination selection without requiring the user to use a personal phone.
- The local browser page provides the same guidance when the user prefers mobile access.
- The master-hotspot approach removes dependence on external internet connectivity.
- BLE UUID detection with RSSI thresholding is adequate for room-level or junction-level route progression, but not for exact location tracking.



Fig. 6. Working display module of the prototype showing the local navigation interface on the 2.4 inch touch LCD shield, including destination status, live routing state, next-instruction guidance, and destination buttons.

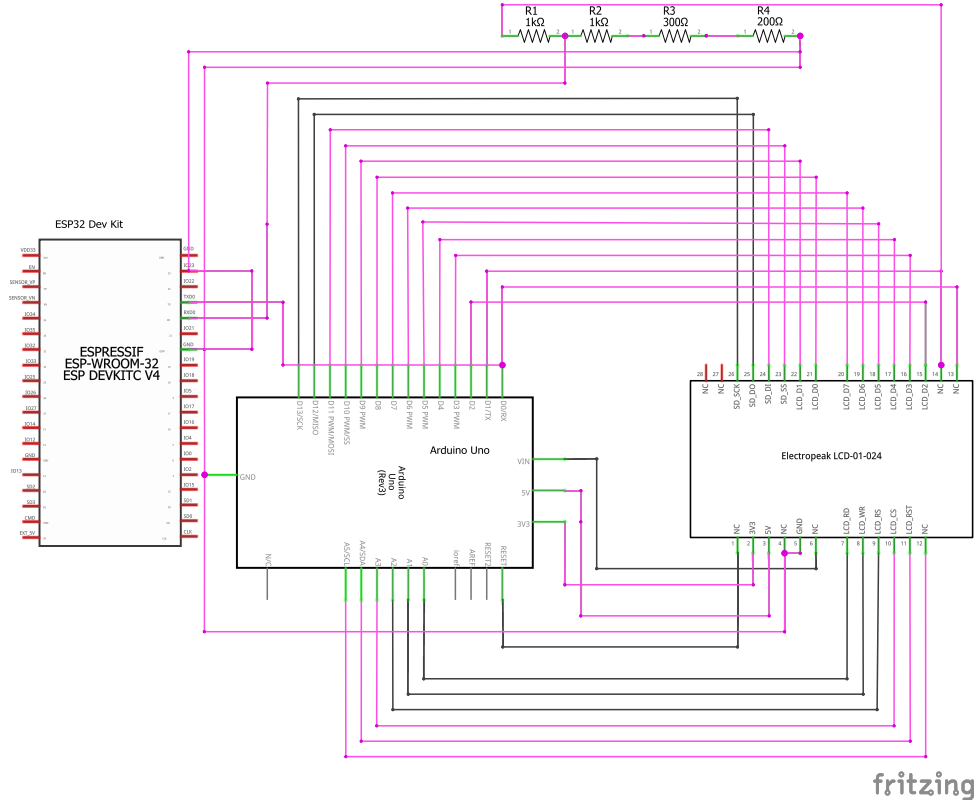


Fig. 4. Schematic diagram of the prototype showing the ESP32, Arduino UNO, LCD module, resistor network, and visible pin connections used for hardware interfacing.

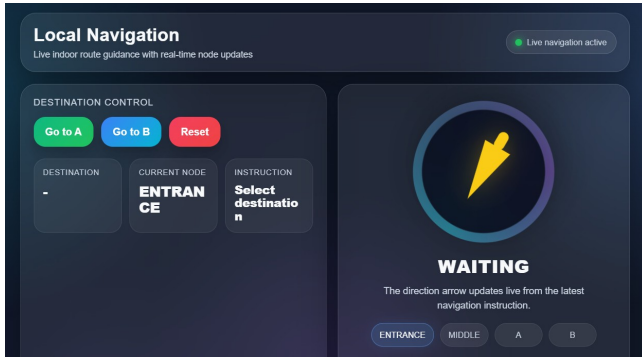


Fig. 7. Web-based navigation dashboard of the prototype showing destination controls, current node, instruction status, direction indicator, and live checkpoint state for local indoor navigation.

XV. EMERGENCY-USE CASE RELEVANCE

The proposed system is especially useful in emergency and urgent-service scenarios. A visitor or patient attendant may need immediate guidance to an emergency room, registration point, help desk, or treatment room. In such conditions, static signboards may not be enough. A checkpoint-based system can give short and direct instructions such as “go to the middle junction” and “turn left to Room A”, thereby reducing confusion and saving time.

XVI. ADVANTAGES

The proposed system offers the following advantages:

- Practical checkpoint-based navigation without full localization complexity.
- Direct destination selection on the touch LCD shield.
- Optional phone-based guidance through a local web page.
- Master-node hotspot operation without dependence on external internet.
- Useful for hospitals, service centers, offices, and academic buildings.
- Suitable for room-level and junction-level navigation.

XVII. LIMITATIONS

Despite its usefulness, the system has certain limitations:

- It does not provide precise real-time indoor coordinates.
- BLE detection can vary due to signal fluctuation and obstacles.
- The current architecture uses ESP32 boards at each node, which increases per-node hardware cost.
- The system is currently more suitable for small single-floor or limited-area buildings.
- Guidance quality depends on correct checkpoint placement and stable local communication.

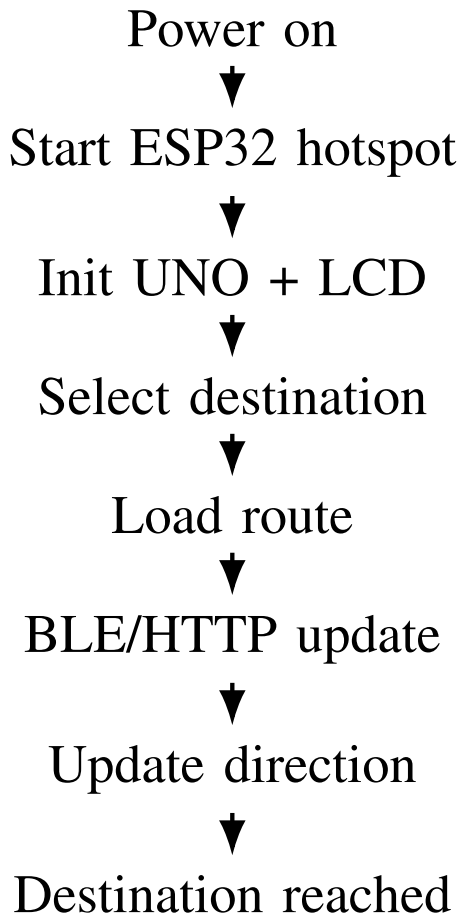


Fig. 5. Software flow of the prototype from hotspot startup to checkpoint-based destination confirmation.

XVIII. FUTURE WORK AND COST REDUCTION OPPORTUNITIES

The current prototype uses ESP32 development boards for all nodes because they simplify prototyping, software development, and integration of Wi-Fi with BLE. However, this also makes the present node architecture more expensive than necessary for large deployment. In future work, the system can be made cheaper in the following ways:

- Keep the entrance unit as the main ESP32 master node, but replace some room checkpoint nodes with simpler and lower-cost hardware.
- Use cheaper endpoint modules such as ESP8266-class Wi-Fi nodes where BLE is not essential.
- Use BLE-only beacon modules or passive tags for fixed room checkpoints while retaining one or two smarter gateway nodes.
- Replace multiple development boards with custom PCBs after the design is finalized.
- Restrict the touch LCD shield and Arduino UNO display unit to the entrance node only and avoid display hardware at intermediate checkpoints.

- Use QR codes or NFC tags for selected low-priority rooms so that not every destination needs an active wireless node.
- Combine battery and enclosure design into a single optimized node package for mass deployment.

Other future extensions include multi-floor support, voice guidance, multilingual instructions, integration with emergency alert routing, and better route analytics using logged movement events.

XIX. CONCLUSION

This paper presented an indoor wayfinding system for small buildings using ESP32 checkpoint nodes, a master hotspot node, an Arduino UNO with a 2.4 inch touch LCD shield, and a local web interface. The system is designed around checkpoint-based navigation rather than full indoor localization. In the current prototype, the user can select destination A or B on the entrance display unit or through the phone-based browser page, and route progression is updated when checkpoint nodes detect a predefined BLE service UUID and report the node to the master over HTTP. The architecture is practical for emergency guidance and small-building navigation. Although the current all-ESP32 checkpoint setup is convenient for prototyping, future versions can reduce deployment cost by introducing mixed hardware nodes and more specialized checkpoint devices.

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